

# STEADY

## Therapeutic Load & Readiness

Implementation-Ready Engineering Handoff

Clinical Intelligence Expansion | Document 05 | Repository-grounded against tprawlins-lang/Emdr | branch  
claude/gifted-keller-501y5d | September 2026

### Build directive

Add a clinician-only decision-support layer that describes how much therapeutic work the person appears to be tolerating and recovering from. It combines safety state, session intensity, delayed recovery, sleep, dissociation, functional progress, Response Fingerprint, and Recovery Trajectory into an evidence-backed recommendation to stabilize, maintain, or consider progression. It never authorizes treatment or overrides safety.

### Shared product rule

The clinician decides what clinical meaning and action to accept. Steady may measure, organize, compare, and suggest, but it does not silently turn an association, model output, or behavioral signal into clinical fact.

# Repository foundation this handoff assumes

| Foundation                  | Current Steady anchor            | Rule for this build                                                                                                                  |
|-----------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Longitudinal history        | src/lib/spine.ts + ADR 0010      | Durable state must be event-backed and replayable. Do not create a parallel mutable history.                                         |
| Tenant/person boundary      | src/lib/repository.ts + ADR 0011 | Every new durable patient record carries tenant_id and person_id; new paths use TenantContext even while legacy paths are migrating. |
| Clinician patient workspace | src/app/clinician/member/[id]/*  | New clinician surfaces extend PersonShell and existing evidence/review conventions.                                                  |
| Evidence-linked summary     | src/lib/clinical/summary.ts      | A generated claim without resolvable evidence is withheld. Summary is a reading aid, not the record.                                 |
| Trajectory                  | src/lib/clinical/trajectory.ts   | Separate scales stay separate; provenance remains visible; do not invent a composite clinical meaning from arbitrary axis alignment. |
| Work queue                  | src/lib/clinical/work-queue.ts   | Clinician home opens on actionable work, stable server ordering, clear reason, one primary action.                                   |
| Companion memory            | src/lib/companion.ts             | Patient/Companion data retains source class and memory exposure rules. AI output is not clinician evidence.                          |
| AI Gateway                  | ADR 0012                         | New model tasks must route through the gateway with schema validation, task version, provenance, evaluation, and PHI policy.         |

This series also assumes the Clinician Thoughts / Clinical Memory / Session Prep Engineering Specification v2.1 is implemented or being implemented as the common clinician-memory and evidence layer.

## Cross-feature invariants

- **Safety authority stays deterministic.** These features may add clinician decision support but cannot clear, weaken, bypass, or replace the safety engine.
- **Stabilization scope stays intact.** No feature autonomously initiates trauma reprocessing, changes a gated treatment decision, diagnoses, or issues a treatment order.
- **Source classes never collapse.** Patient report, clinician observation, system measurement, formal note, Companion interaction, and model inference remain distinguishable.
- **Associations are associations.** A temporal or statistical relationship is not described as causation unless a clinician explicitly records that judgement from appropriate evidence.
- **No future-data leakage.** Historical reconstruction and point-in-time evaluations use only information Steady knew at the relevant cutoff.
- **Missing data stays visible.** Missing observations do not become zero, normal, compliant, or recovered.
- **Corrections append.** An authorized correction supersedes prior derived state without erasing history.
- **Model output is disposable; evidence and provenance are permanent.** Every material intelligence output points to evidence IDs and task/policy versions.

## 1. Authority boundary

**Safety asks: is the person allowed to access this activity under Steady's deterministic rules?**

**Therapeutic Load asks: based on observed recovery, how much additional intensity appears prudent for a clinician to consider?**

The second can never answer the first or override it.

- This is clinician decision support, not autonomous treatment sequencing.
- In the stabilization/resourcing deployment scope, this engine cannot initiate, unlock, schedule, or recommend autonomous trauma reprocessing.

- A "consider progression" recommendation means "there is enough favorable evidence for a clinician to review whether the next step is appropriate," not "progress the patient."
- If the safety engine blocks an activity, Therapeutic Load displays that external constraint and stops. It does not compute a workaround.

## 2. Inputs

| Input class                | Examples                                                       | Role                                                                      |
|----------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| deterministic safety state | hard stop, unsafe check-in, crisis route, gate decision        | External hard constraint; never reinterpreted.                            |
| session intensity          | pre/peak/post distress, session duration, hard stop, module    | Load exposure.                                                            |
| delayed recovery           | post-session recovery confirmation, next-day activation, sleep | Recovery burden.                                                          |
| Response Fingerprint       | observed response and recovery burden by intervention          | Personal tolerance/context.                                               |
| Recovery Trajectory        | function, sleep, symptoms, recovery domains                    | Whether broader course is improving/stable/reversing.                     |
| Return-to-Life             | functional movement toward meaningful goals                    | Capacity in real life.                                                    |
| dissociation / activation  | check-in state and change                                      | Current regulation context.                                               |
| clinician judgement        | Thoughts, explicit readiness statement, follow-up              | Human context may outweigh model synthesis; preserved as clinician input. |

## 3. Recommendation states

| State                | Meaning                                                                                                           | System action                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| blocked_by_safety    | Existing safety/gate system prevents relevant progression                                                         | Display reason and safer existing path; no load recommendation beyond that. |
| insufficient_data    | Not enough recovery evidence                                                                                      | Ask clinician to rely on judgement; state missing evidence.                 |
| stabilize            | Repeated recovery burden or broader deterioration suggests reducing additional load and emphasizing stabilization | Decision support only. No automatic plan change.                            |
| maintain             | Current load appears tolerated; evidence does not justify a stronger progression suggestion                       | Decision support only.                                                      |
| consider_progression | Repeated recovery is favorable and functional/trajectory evidence is not deteriorating                            | Invite clinician review. Never auto-unlock or initiate treatment.           |

## 4. Therapeutic load dimensions

| Dimension               | Example evidence                           | High-load / low-capacity indication                  |
|-------------------------|--------------------------------------------|------------------------------------------------------|
| acute load              | peak distress, high session activation     | Repeated high peaks or hard stop.                    |
| recovery time           | hours/days to baseline, post-session check | Longer or unresolved recovery.                       |
| sleep cost              | sleep-quality change after work            | Repeated post-session sleep worsening.               |
| dissociation cost       | post-session or next-day dissociation      | Repeated increase.                                   |
| functional cost/benefit | Return-to-Life movement                    | Function stalls/reverses despite repeated high load. |
| intervention tolerance  | Response Fingerprint                       | Mixed/adverse pattern under current context.         |
| trajectory capacity     | Recovery Trajectory                        | Multiple corroborating domains reversing/stalling.   |

## 5. Database schema

```
CREATE TABLE therapeutic_load_snapshots (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
```

```

state TEXT NOT NULL CHECK (state IN (
    'blocked_by_safety', 'insufficient_data', 'stabilize', 'maintain', 'consider_progression'
)),
policy_version TEXT NOT NULL,
evidence_cutoff TEXT NOT NULL,
load_dimensions_json TEXT NOT NULL,
capacity_dimensions_json TEXT NOT NULL,
safety_constraint_ref TEXT,
explanation_json TEXT NOT NULL,
limitations_json TEXT NOT NULL DEFAULT '[]',
computed_at TEXT NOT NULL
);

CREATE TABLE therapeutic_load_evidence (
    snapshot_id TEXT NOT NULL,
    evidence_type TEXT NOT NULL,
    evidence_id TEXT NOT NULL,
    role TEXT NOT NULL CHECK (role IN ('load', 'capacity', 'constraint', 'context')),
    PRIMARY KEY(snapshot_id, evidence_type, evidence_id, role)
);

CREATE TABLE therapeutic_load_reviews (
    id TEXT PRIMARY KEY,
    tenant_id TEXT NOT NULL,
    person_id TEXT NOT NULL,
    snapshot_id TEXT NOT NULL,
    clinician_person_id TEXT NOT NULL,
    decision TEXT NOT NULL CHECK (decision IN (
        'acknowledged', 'agree_stabilize', 'agree_maintain', 'review_progression', 'disagree', 'defer'
    )),
    note TEXT,
    created_at TEXT NOT NULL
);

```

## 6. Deterministic policy engine

1. READ existing safety/gate decision.  
if blocked -> state = blocked\_by\_safety; STOP.
  2. REQUIRE minimum recovery evidence.  
if insufficient -> insufficient\_data.
  3. ASSESS recovery burden across repeated recent work:
    - unresolved post-session recovery
    - longer recovery time than prior pattern
    - repeated sleep/dissociation worsening
    - hard-stop / adverse Response Fingerprint evidence
  4. ASSESS capacity:
    - stable or improving Return-to-Life function
    - favorable/stable recovery trajectory
    - shorter/consistent recovery
    - sufficient stabilizing intervention response
  5. APPLY versioned conservative policy:
    - repeated burden + weak capacity -> stabilize
    - mixed / tolerated / unclear -> maintain
    - repeated favorable recovery + no deterioration -> consider\_progression
  6. EMIT evidence and limitations.
- No model decides the state.

## 7. Conservative default rules

- No "consider progression" from one good session.
- Any unresolved existing safety state prevents progression suggestion.
- Repeated delayed recovery can outweigh immediate distress reduction.
- Functional deterioration or multi-domain trajectory reversal prevents a favorable progression suggestion until clinician review, even if immediate session SUDS decreases.

- Missing delayed follow-up prevents the system from assuming good recovery.
- Clinician explicit uncertainty from Thoughts appears as context and can force "maintain" or "defer" display policy if configured, but should not be silently treated as a deterministic safety rule.

## 8. UX

- **Session Prep:** "Load & recovery" section with state, 2-4 evidence bullets, and Why am I seeing this?.
- **Patient overview:** clinician-only card. Do not show a mysterious readiness score.
- **Detail:** load dimensions, capacity dimensions, safety constraint if any, recent recovery timeline, Response Fingerprint links, Return-to-Life links, trajectory links, limitations.
- **Clinician action:** Acknowledge, Agree stabilize, Maintain, Review progression, Disagree, Defer. These actions record judgement; they do not automatically change plan/gates.

## 9. Command Center integration

- Use Handoff 03 provider contract.
- Create a review\_today signal for a new persistent stabilize state when not already represented by a safety alert.
- Create a follow\_up/review signal for a new consider\_progression state only when clinician policy says a review is useful. Do not nag on every recomputation.
- Deduplicate against existing safety hard stops and unresolved post-session alerts.

## 10. AI role

| Task                                 | Allowed                                                                                  | Forbidden                                                |
|--------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------|
| therapeutic_load.explain             | Translate deterministic dimensions/state into concise evidence-linked clinician language | Change state, safety gate, plan, or threshold.           |
| therapeutic_load.retrieve_context    | Find relevant clinician Thoughts/threads for context                                     | Treat model-selected context as a hard constraint.       |
| therapeutic_load.session_prep_render | Format recommendation and evidence for prep                                              | Tell clinician to initiate autonomous trauma processing. |

## 11. Events

| Event                                       | Meaning                                 |
|---------------------------------------------|-----------------------------------------|
| therapeutic_load.snapshot_computed.v1       | Versioned recommendation snapshot.      |
| therapeutic_load.recommendation_proposed.v1 | New clinician-visible state/change.     |
| therapeutic_load.clinician_reviewed.v1      | Human decision recorded.                |
| therapeutic_load.recommendation_changed.v1  | State changes with new evidence/policy. |

## 12. Implementation phases

### Phase 1 - Input adapters + safety boundary

#### Build

- safety/gate adapter
- session load/recovery adapter
- Return-to-Life/Response/Trajectory adapters

#### **Definition of done**

- Blocked safety state short-circuits calculation
- point-in-time cutoff honored

### **Phase 2 - Conservative policy engine**

#### **Build**

- dimension calculations
- minimum evidence policy
- state machine
- limitations

#### **Definition of done**

- No model state decision
- one good session cannot produce progression

### **Phase 3 - Clinician UX + reviews**

#### **Build**

- Session Prep card
- patient detail
- review actions

#### **Definition of done**

- Human review is auditable and does not auto-change plan

### **Phase 4 - Command Center provider**

#### **Build**

- stabilize/review-progression signals
- dedupe against safety

#### **Definition of done**

- No duplicate safety work
- repeated recompute does not spam queue

## **13. Acceptance criteria**

- Safety gate is read first and always outranks therapeutic-load output.
- Repeated delayed recovery can produce stabilize even when immediate session response looks favorable.
- Return-to-Life and Recovery Trajectory evidence can prevent a misleading progression suggestion.
- No readiness number is displayed without explanation; preferred design is categorical evidence-backed state.
- Clinician disagreement is recorded and does not erase system evidence.
- No system action autonomously changes treatment intensity, module access, or trauma-processing status.

## Appendix A. File-level implementation map

| Path                                                     | Change                                                                               |
|----------------------------------------------------------|--------------------------------------------------------------------------------------|
| src/lib/clinical/therapeutic-load.ts                     | NEW policy/state engine.                                                             |
| src/lib/clinical/therapeutic-load-policy.ts              | NEW versioned conservative thresholds.                                               |
| src/lib/clinical/therapeutic-load-evidence.ts            | NEW adapters to safety, sessions, goals, response, trajectory.                       |
| src/components/clinical/TherapeuticLoadCard.tsx          | NEW clinician/session-prep card.                                                     |
| src/app/clinician/member/[id]/load/page.tsx              | NEW detail/review surface or integrate into existing plan/progress IA.               |
| src/lib/clinical/attention-providers/therapeutic-load.ts | NEW Handoff 03 provider.                                                             |
| src/lib/spine.ts + projections.ts                        | Add snapshot/review events.                                                          |
| src/lib/db.ts + scripts/pg-schema.sql                    | Add/mirror schema.                                                                   |
| src/lib/ai-gateway/*                                     | Register explanation/context tasks.                                                  |
| tests/therapeutic-load*.test.ts                          | Safety dominance, missing recovery, mixed response, security, replay, point-in-time. |